CSE231 Operating System (Section A), Quiz 01 Monsoon 2025
Time allocated: 4pm – 4:20pm

Name	
Roll Number	

Instructions:

- This **is a closed book and closed notes quiz. Please be aware of strict plagiarism policy.**
- For questions requiring justification, please be as concise as possible. 2-3 sentences would be the ideal size of a justification. No extra pages will be provided.
- **NO EXTRA ANSWER SHEET WILL BE PROVIDED. USE THE GIVEN SPACE WISELY**

Question-1: Arrange the following stages/output of GCC compilation in the correct order of code generation: a) linking with glibc, b) relocatable file, c) assembly code, d) preprocessed code, and e) a.out. No justifications needed. Only provide the option numbers in correctly arranged order. **[1 marks]**

Question-2: Provide the GCC flag **(if any)** that would be needed for the below-mentioned stages/output starting from the input file hello.c. No justifications needed. **[2 marks]**

a) Linking with glibc: ____________________ b) Relocatable file: ______________________

c) Assembly code: _______________________ d) a.out: ____________________

Question-3: The job of a **Cook** is to do **Cooking** using a **Recipe** and managing the **Kitchen**. In this analogy, if the **Cook** points to the role of an **Operating System** and **Kitchen** relates to the **computer system**, then what are the two things that the Recipe contains (from the OS point of view)? No justifications needed. **[1 marks]**

Question-4: Name one disadvantage of dynamic linking. **[1 marks]**

Question-5: Using the information provided in the Elf header (see Figure-1), write the formula to calculate the byte offset (from starting of the file) to the **LAST** program header in Program Header Table (PHT). **[2 marks]**

```
typedef struct {
        unsigned char   e_ident[EI_NIDENT];
        Elf32_Half      e_type;
        Elf32_Half      e_machine;
        Elf32_Word      e_version;
        Elf32_Addr      e_entry;
        Elf32_Off       e_phoff;
        Elf32_Off       e_shoff;
        Elf32_Word      e_flags;
        Elf32_Half      e_ehsize;
        Elf32_Half      e_phentsize;
        Elf32_Half      e_phnum;
        Elf32_Half      e_shentsize;
        Elf32_Half      e_shnum;
        Elf32_Half      e_shstrndx;
} Elf32_Ehdr;
```

Figure-1

Question-6: What is the first ELF section that would be looked up by the OS for executing an executable file. No justifications needed. **[1 marks]**

Question-7: What happens to EBP and ESP during a method's a) epilogue and b) prologue. One line summary for each of the two sub-questions **[2 marks]**

Answer-1: d-c-b-a-e **[No partial marking]**
Answer-2: a) **no flag needed**, b) "**-c**", c) "**-S**", d) **no flag needed**. **[Marking: 0.5 x 4]**
Answer-3: a) instructions, and b) program parameter/variables. We will provide marks if you have mentioned about contents of ELF file (e.g., section header / sections, etc.). **[Marking: 0.5 x 2]**
Answer-4: See slide #20 Lecture 02. **[No partial marking]**
Answer-5: e_phoff + { (e_phnum – 1) x e_phentsize) } **[No partial marking. If there is no "-1" then +1 only]**
Answer-6: Interp (we will provide marks if you have mentioned .text section also in place of .interp)
Answer-7: Prologue: Old EBP value is saved in stack slot currently pointed by ESP **[+0.5 marks]** and EBP is updated to point to this new location **[+0.5 marks]**.
Epilogue: Stack slot storing the old EBP value is popped and loaded into EBP register **[+0.5 marks]** and ESP is adjusted/moved to the point to the last slot in the stack frame of the caller. **[+0.5 marks]**